



Semester 1 Examinations 2014/2015

Exam Code(s)	1SD1, 1MF1, 1SD3
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems
Paper No.	I
Repeat Paper	No
External Examiner(s)	Professor Liam Maguire
Internal Examiner(s)	Professor Gerard Lyons *Dr. Petronel Bigioi *Dr. Enda Howley *Dr. Matthias Nickles

Instructions: **CT861 and CT516:** Answer **two** questions from Section A and **one** question from Section C.

CT538 and CT542: Answer **two** questions from Section A and **one** question from Section B.

Use a separate answer book for each section!

Duration	2 hrs
No. of Pages	5
Discipline(s)	Information Technology

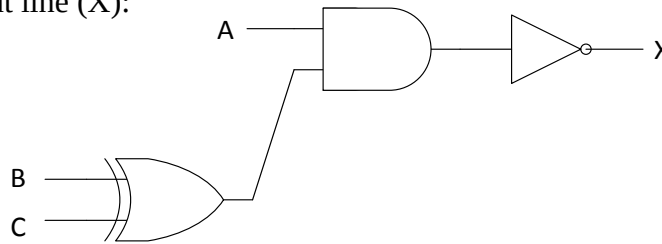
Requirements	None
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Release to Library: Yes X No

Section A

Question 1

- a) Name and explain the benefits of *dual core* microprocessors over *single core* microprocessors. [5 MARKS]
- b) You are given the following circuit with three input lines (A, B, C) and one output line (X):



Provide the *truth table* for this circuit.

(Remark: no need to provide Boolean expressions.)

[8 MARKS]

- c) Describe three different techniques for *memory management*. Also describe advantages and disadvantages of each of these techniques. [12 MARKS]

Question 2

- a) Name three different types of (CPU-) *instructions* and describe their purposes. [6 MARKS]
- b) What are the steps required to obtain the *Two's Complement* representation of an integer number? What is the Two's Complement of the 6-bit binary number 010101? [9 MARKS]
- c) Explain the *instruction cycle with interrupts*. Use a diagram to illustrate your explanation. [10 MARKS]

Question 3

- a) What is a *logic gate*? Provide an example to illustrate your explanation. [7 MARKS]
- b) Explain what a *multiplexer* is. In addition, provide a truth table for a multiplexer with four input lines and one output line. [10 MARKS]
- c) What is *cache memory*, and what is it used for? Also, describe the steps of a cache operation. [8 MARKS]

PTO

Question 4

- a) Describe the technique of *paging* (in the context of memory management). What drawbacks does paging have? [7 MARKS]
- b) Describe *programmed I/O* and *interrupt-driven I/O*. Which one is typically preferred in modern computers, and why? [8 MARKS]
- c) Describe the *five-state process model*. Provide a diagram in support of your explanation. [10 MARKS]

PTO

Section B

Question 5

Describe the TCP/IP Reference model. Describe briefly each layer. Show how the TCP/IP layers map onto OSI reference layers.

[25 MARKS]

Question 6

Describe CSMA/CD protocol. Why is there a minimum frame length required by Ethernet standards? Describe the binary exponential back-off algorithm and why it is useful for Ethernet.

[25 MARKS]

PTO

Section C

Question 7

- a) Explain the concept of *critical sections* (in the context of process interaction).
Provide an example of your choice to illustrate your explanation. [9 MARKS]
- b) Explain the concept of *semaphores* and provide pseudo-code (or real code) to illustrate your explanation. [9 MARKS]
- c) What does *starvation* mean (in the context of process interaction)? How does starvation manifest itself if three processes compete for one resource? [7 MARKS]

Question 8

- a) What is the purpose of a *disk cache* and how does it work? [7 MARKS]
- b) Explain *contiguous allocation*. Provide an example *FAT* (of your own choice) to illustrate your explanation. Also, name and describe a particular problem which can occur with contiguous allocation. How can this problem be addressed? [10 MARKS]
- c) Describe the role and tasks of a *device manager* (in the context of operating systems). [8 MARKS]